Reg. No.	R	A	2	3	1	1	0	0	3	0	1	1	7	6	1

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

First Semester

21MAB101T - CALCULUS AND LINEAR ALGEBRA

(For the candidates admitted from the academic year2021-2022 to 2023 - 2024)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer **ALL** Questions

		Marks	BL	CO	PO
1.	The eigen values of the matrix $\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ are A) 2, 3 B) -2, -3 C) 6, -1 D) -6, 1	1	2	1	2
2.	The canonical form $-y_1^2 + 3\,y_2^2 + \,y_3^2$ has the index A) 2 B) 3 C) 1 D) 0	1	1	1	2
3.	If $A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 4 \end{bmatrix}$, then the eigen values of A^2 are A) 3, 2, 4 B) 1/3, 1/2, 1/4 C) 9, 4, 16 D) 27, 8, 64	1	2	1	2
4.	The matrix of the quadratic form $x_1^2 + x_2^2 - 3x_3^2 - 2x_1x_2 + 4x_1x_3$ is A) $\begin{bmatrix} 1 & 1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & -3 \end{bmatrix}$ B) $\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 0 \\ -2 & 0 & -3 \end{bmatrix}$ C) $\begin{bmatrix} 1 & -1 & 2 \\ 1 & -1 & 0 \\ 2 & 0 & -3 \end{bmatrix}$ D) $\begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & -3 \end{bmatrix}$	1	1	1	2
5.	If $z = x^3 + 3x^2y + y^3$, then $\frac{\partial z}{\partial x} =$ A) $3x^3 + 6x$ B) $3x^2 + 6xy$ C) $3x - 6xy$ D) $3x^2 - 6xy$	1	1	2	2
6.	If $f(x, y) = 0$ and y is an implicit function of x, then $\frac{dy}{dx}$ is A) $-\frac{f_x}{f_y}$ B) $-\frac{f_y}{f_x}$ C) $\frac{f_x}{f_y}$ D) $\frac{f_y}{f_x}$	1	1	2	2
7.	If u and v are fucntionally dependent then their Jacobian value is A) 1 B) 2 C) 0 D) 3	1	1	2	2

8. If $x = r\cos\theta, y = r\sin\theta$, then $J\left(\frac{x,y}{r,\theta}\right)$ is — 1 1 2 2

A) r/2 B) 2/r
C) 1/r D) r

9. The complementary function of $(D^2 - 4D + 4)y = 4$ is — 1 2 3 2

A) $(C_1 + C_2x)e^{2x}$ B) $(C_1 + C_2x)e^{x/2}$
C) 0 D) $C_1x + C_2x^2$

10. The particular integral of $(D^2 + 16)y = e^{-3x}$ is — 1 2 3 2

A) $\frac{e^{-3x}}{16}$ B) e^{-3x}
C) $\frac{e^{3x}}{25}$ D) $\frac{e^{-3x}}{25}$

11. The complementary function of $(x^2D^2 - xD + 1)y = 0$ is — 1 2 3 2

A) $C_1 + C_2\log x$ B) $C_1 + C_2x$
C) $(C_1 + C_2\log x)x$ D) $(C_1 - C_2\log x)x$

12. The Legendre's equation $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 0$ takes the form — 1 2 3 2

A) $(\theta^2 + 1)y = 0$ B) $(\theta + 1)y = 0$
C) $(\theta^3 + 1)y = 0$ D) $(\theta - 1)y = 0$

13. $x = at^2,\ y = 2at$ is the parametric coordinates of — 1 1 4 2

A) Parabola B) Rectangular Hyperbola
C) Hyperbola D) Ellipse

14. The reciprocal of the curvature of the curve at any point 'P' is called — 1 1 4 2

A) Centre of Curvature B) Circle of Curvature
C) Chord of Curvature D) Radius of curvature

15. The Locus of centre of curvature is called — 1 1 4 2

A) Evolute B) Involute
C) Envelope D) Curvature

16. The envelope of the family of straight lines $y = mx + \frac{1}{m}$ is — 1 2 4 2

A) $x^2 = 4y$ B) $y^2 = 4x$
C) $y^2 = 4x^2$ D) $x = y$

17. The P series $\frac{1}{1^P} + \frac{1}{2^P} + \ldots\ldots + \frac{1}{n^P}$, it converges, if — 1 1 5 2

A) $P = 1$ B) $P < 1$
C) $P > 1$ D) $P < -1$

18. Let $\sum U_n$ be a convergent series of positive terms and let $\sum V_n$ be a divergent series of the positive terms then — 1 1 5 2

A) The sequence $\{V_n\}$ diverges to ∞ B) The sequence $\{U_n\}$ converges to 0
C) The sequence $\{U_n\}$ is convergent and the sequence $\{V_n\}$ is divergent D) Both the sequence $\{U_n\}$ and $\{V_n\}$ are convergent

19. The series $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\ldots \infty$ represent the expansion of — 1 1 5 2

A) e^x B) $\sin x$
C) $\cos x$ D) $\log(1+x)$

20. The Series $\frac{1+n}{1+n^2}$ is — 1 2 5 2

A) Convergent B) Divergent
C) Oscillatory D) Constant

PART - B (5 x 8 = 40 Marks)
Answer **ALL** Questions

		Marks	BL	CO	PO
21 a.	Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	8	3	1	2
	(OR)				
b.	By using Cayley-Hamilton theorem, find A^4 of the matrix $A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$	8	3	1	2
22 a.	Find the Taylor's series expansion of $e^x \sin y$ near the point $\left(-1, \frac{\pi}{4}\right)$ upto second degree terms.	8	3	2	2
	(OR)				
b.	Examine the function $f(x, y) = x^3y^2(12 - x - y)$ for extreme values.	8	3	2	2
23 a.	Solve by method of variation of parameters $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$.	8	3	3	2
	(OR)				
b.	Solve $(x^2D^2 - 3xD + 4)y = x^2$	8	3	3	2
24 a.	Find the equation of circle of curvature of $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point $\left(\frac{a}{4}, \frac{a}{4}\right)$	8	3	4	2
	(OR)				
b.	Find the equation of the evolute of the parabola $x^2 = 4ay$	8	3	4	2
25 a.	Discuss the convergence of the series $\sum_{n=1}^{\infty} \frac{n^3+1}{2^n+1}$	8	3	5	2
	(OR)				
b.	Test the convergence of the series $x + \frac{1}{2}\frac{x^3}{3} + \frac{1.3}{2.4}\frac{x^5}{5} + \frac{1.3.5}{2.4.6}\frac{x^7}{7} + \ldots\ldots\infty,\ x > 0$	8	3	5	2

PART - C (1 x 15 = 15 Marks)
Answer **ANY ONE** Question

		Marks	BL	CO	PO
26.	Reduce the quadratic form $x_1^2 + 2x_2^2 + x_3^2 - 2x_1x_2 + 2x_2x_3$ to canonical form by an orthogonal reduction. Also find the nature of the quadratic form and rank, index, signature of the QF.	15	4	1	2
27.	Find the shortest and longest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$, using Lagrange's method of constrained maxima and minima.	15	4	2	2

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